A New Hopfield-type Neural Network Approach to Multi-Goal Vehicle Navigation in Unknown Environments

Chaomin Luo, Jing Zhang, Gene Eu Jan and Hongwei Mo

***Abstract* — A Hopfield-type neural networks (HNN) algorithm associated with histogram navigation method is proposed in this paper for real-time map building and path planning for *multiple goals* applications. In real world applications such as rescue robots, service robots, mining mobile robots, and mine searching robots, etc., an autonomous vehicle needs to reach multiple goals with a shortest path that, in this paper, is capable of being implemented by a HNN method with minimized overall distance. Once a global trajectory is planned, a foraging-enabled trail is created to guide the vehicle to the multiple goals. A histogram-based local navigation algorithm is employed to plan a collision-free path along the trail planned by the global path planner. A re-planning-based algorithm aims to generate trajectory while an autonomous vehicle explores through a terrain with map building in unknown environments. In this paper, simulation and experimental results demonstrate that the real-time concurrent mapping and multi-goal navigation of an autonomous vehicle is successfully performed under *unknown* environments.**

Keywords* — *Hopfield-type neural networks; Multi-goal motion planning; Autonomous vehicle; Mapping; Unknown Environments

I. INTRODUCTION

MOTION planning and mapping of autonomous vehicles is an important issue. Concurrent motion planning and mapping of robotics is to search a suitable collision-free path of an autonomous vehicle to move from an initial position to a goal designation while the vehicle builds up a map, in a unknown environment. Research on such point-to-point vehicle motion planning has been well carried out recently, indeed, which is a sort of single-goal navigation. In real-time world applications, such as rescue robots, service robots, mine rescue robots, and mine searching robots, etc., **multi-goal** motion planning of autonomous vehicles is highly desirable. In addition to the service robots, many other robotic applications require multi-goal navigation such as vacuum robots, land mine detectors, lawn mowers, and windows cleaners. The multi-goal path planning aims to search a collision-free path for visiting a sequence of *goals* with the minimized total route under unknown environments.

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There have been a large number approaches proposed on autonomous vehicle motion planning ([1-7]). Gu *et al.* [1] proposed an elastic-band-based motion planning model, in which tunability and stability are focused on the vehicle navigation. Raja *et al.* [2] proposed an introducing a gradient function in the conventional potential field method associated with a GA-based method for vehicle motion planning. Besides an attractive force, a repulsive force, a tangential force and a gradient force are introduced in the conventional potential field method integrated with GA path planning model. Davies and Jnifene [3] developed a Genetic Algorithm (GA) path planner to guide an autonomous vehicle to reach specified multiple goals with obstacle avoidance. However, a couple of *artificial* waypoints should necessarily be predefined to assist in preventing the vehicle from the deadlock or escaping from the local minima, which causes more timing delay, more length and more cost.

Luo and Yang [4-5] developed a bio-inspired neural network model that concurrently performs mapping and path planning tasks. Yang and Meng [6] proposed a biologically inspired neural network approach to real-time collision-free motion planning of vehicles in a non-stationary environment. Each neuron in the topologically organized neural network has only local connections. The real-time vehicle motion is planned through the dynamic activity landscape of the neural network. However, multi-goal navigation studies have not been performed largely for intelligent vehicle systems. Faigl and Macak [7] developed a self-organizing map method in conjunction with an artificial potential field based navigation function to generate an optimal path of an autonomous vehicle to visit multiple goals, through a Traveling Salesman Problem (TSP) tool. The autonomous vehicle visits multiple targets just as the traveling salesman problem but with the presence of obstacles, in which mobile vehicles are navigated with the discrete map representation filled with some exact cell in workspace. Although a multi-goal navigation model of an autonomous vehicle modeled as a point vehicle is implemented by Gopalakrishnan, and Ramakrishnan [8], the model lacks of map building component, neither.

In this paper, a real-time concurrent multi-goal motion planning and map building approach of an autonomous vehicle in completely **unknown** environments is proposed. A local map composed of cells is dynamically built through the histogram-driven local navigator with restricted sensory information such as LIDAR-based data while it is navigated under completely unknown environments with obstacle avoidance to visit multiple requested goals. The sensory information obtained by onboard sensors mounted on the vehicle is utilized for its real-time concurrent multi-goal navigation and map building in unknown sceneries.

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The rest of this paper is organized as follows. The well-known D*-Lite global path planner and a Modified VFH local navigator are addressed in Section. In Section III, the collision-free multi-goal navigation of the autonomous mobile vehicle by HNN-based TSP is presented. The map representations of the environments and map building, and the real-time concurrent map building and collision-free multi-goal navigation of the autonomous vehicle are reported in Section IV. Afterwards, the simulation studies and experimental test of the real-time multi-goal navigation and map building of the autonomous vehicle in various unknown environments are described in Section V. Finally, several important properties of the proposed approach are concluded in Section VI.

II. HISTOGRAM -BASED LOCAL NAVIGATOR AND CELL-BASED MAP BUILDING

The navigation system consists of two layers, one is a D*-Lite global path planner with replanning function and the other is a histogram-based local navigator.

Flexibility and efficiency motivate D*-Lite to be adapted to multi-goal path planning. D*-Lite is an incremental heuristic search algorithm extended from A* algorithm by re-utilizing previous search effort in subsequent search iterations for efficient replanning. D* Lite is based on Lifelong Planning A* introduced by Koenig and Likhachev ([9]), which is an incremental replanning version of A*. A* utilizes a best-first search from a starting point ***S***_start to the goal ***S***_goal guided by the heuristic *h* thus it is able to search an efficiently traversable path between points commonly employed for robotics path planning in known 2D gird-based maps ([10]). The primary principle of D*-Lite in terms of incremental search is to re-utilize previous search effort in the subsequent search iterations, which represents an incremental heuristic replanning version of A*. The incremental heuristic search based D*-Lite algorithm is therefore orders of magnitude more than efficient than A* that plans a new path from scratch repeatedly. In comparison with A*, computational effort of D*-Lite with regard to search time may decrement by a factor of one to two orders of magnitude ([10]). In this paper, the D*-Lite is employed to generate the global trajectory of an autonomous vehicle under ***unknown*** environments.

The local navigator aims to generate velocity commands for the autonomous mobile vehicle to move towards a goal. The inclusion of a sequence of markers in the motion planning, which decomposes the global trajectory generated by the global planner into a sequence of segments, makes the model especially efficient for the workspace densely populated by obstacles. Ulrich and Borenstein [11] first successfully proposed a Vector Field Histogram (VFH) methodology for navigation. The Virtual Force Field (VFF) approach was initially inspired by potential field method in conjunction with the concept of certainty cells. Due largely to failure of avoiding the inherent defects of potential fields, its forward version, VFH method was developed to overcome the shortcomings of VFF ([11]). Afterwards, the enhanced version of VFH approach, namely, VFH, was introduced to take the dimension of vehicle into account. In this paper, VFH is utilized as our LIDAR-based local navigator.

2D cell-based map filled with equally-sized cells, which are marked as either occupied or free, is built as the mobile vehicle moves. Concurrent map building and navigation are the essence of successful multi-goal navigation under unknown environments. Map building is a fundamental task in order to achieve high levels of autonomy and robustness in multi-goal navigation that makes it possible for autonomous vehicles to make decision in positioning with obstacle avoidance.

It is especially beneficial for autonomous vehicles to perform robust multi-goal navigation in unknown terrains, given the fact that it facilitates the utilization of path planning algorithms to determine the optimal trajectory among waypoints as multiple goals. Precise estimate of the vehicle pose (X, Y, Yaw) is demanded by map building so that accurate registration of the local map on the global map is capable of being carried out. In addition, a Kalman Filter is implemented to estimate vehicle pose reliably by fusing data from the motor encoders, the DGPS, and the digital compass. This map building aims to construct a probabilistic-occupancy-cell-based map. The values for each cell in the map vary over the range [-127, 128]. The initial value is zero, which indicates that the cell is neither occupied nor unoccupied. The value is 128 if one cell is occupied with certainty and -127 if one cell is unoccupied with certainty. The values falling into (-127, 128) express contain level of certainty in between.

III. HNN-BASED MULTI-GOAL VISIT

In real world applications of multi-goal navigation and mapping, multiple goals as a sequence of waypoints and a relative cost for travelling between each goal to each other are provided. The objective is to search a route through all the waypoints in which all waypoints are visited once, and to find the shortest overall tour. In rescue vehicle application, for instance, the vehicle starts at one designated waypoint, visits each other waypoint and then ends at the initial waypoint. Traveling Salesman Problem (TSP) is an optimization problem to minimize the travelling distance in a finite number of cities while the cost of travel between each city is known. The classic TSP is employed to deal with the multi-goal visit problem, in which a sequence of goals is visited so that the total planned length of the route is minimized. The objective of this TSP with regard to multi-goal navigation is to search an ordered set of all the waypoints for the autonomous vehicle to visit at such that the cost is minimized. A list of waypoints and distances, or cost, between each of them is necessary for TSP. Therefore, goals are called waypoints with GPS coordinates of the goals in latitude and longitude. The multi-goal visit problem has practical applications on some fields such as transportation plan issues where deliveries are required as well as fuel cost and time are to be minimized.

In this paper, a Hopfield-type neural networks (HNN) model is utilized to resolve the TSP for the multi-goal navigation with multiple waypoints ([12-13]) programmed in MATLAB. HNN algorithm is a form of recurrent neural network served as content-addressable memory systems with

binary threshold nodes. The energy function ***E*** of this system varies with time in accordance with the objective function that is the minimized route distance.

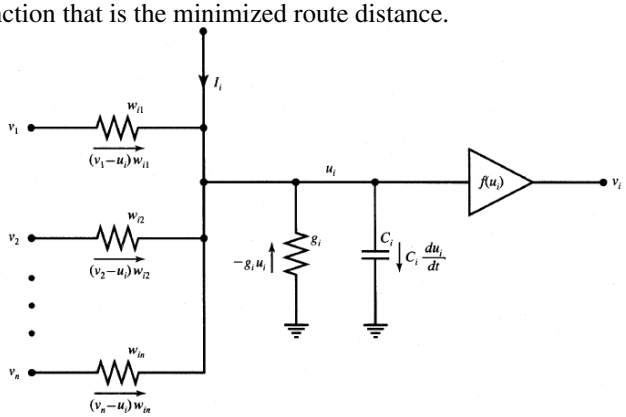


Fig.1 The Hopfield neural network model using electrical components with its input node of the *i*'th neuron (redrawn from Zurada, 1992)

The suitable energy function for the system in Figure 1 is obtained in the form:

$$E(v) = -\frac{1}{2}v^t W v - i^t v + \sum_{i=1}^{n} G_i \int_0^{v_i} f_i^{-1}(z)dz \qquad (1)$$

where, $f_i^{-1}(z)$ represents the inverse of the activation functions f_i Note that the time argument is omitted in Eq. (1). The designed neural system specifications are known in *n*-dimensional output space v^n. Conductance W_{ij} connects the input of the *i*'th neuron to the output of the *j*'th neuron. The total conductance connected to the input node *i* of the neuron is defined as G_i. The current entering the circuit in Fig. 1 is defines as i^t.

Therefore, the energy function of the Hopfield-type network is defined as follows.

$$E = \frac{A}{2}\sum_{x=1}^{N}\left(\sum_{i=1}^{N} v_{xi} - 1\right)^2 + \frac{A}{2}\sum_{i=1}^{N}\left(\sum_{x=1}^{N} v_{xi} - 1\right)^2 + E_0 \quad (2)$$

$$E_0 = \frac{D}{2}\sum_{x=1}^{N}\sum_{y=1}^{N}\sum_{i=1}^{N} v_{xi} d_{xy} v_{y,i+1} \quad (3)$$

The dynamic equation of the continuous Hopfield network is given by

$$\frac{dE(t)}{dt} = -\sum_{j=1}^{n}\frac{dv_j(t)}{dt} \times C_j \frac{dU_j(t)}{dt} \qquad (4)$$

$$\frac{dU_{xi}}{dt} = -\frac{\partial E}{\partial v_{xi}} = -A\left(\sum_{i=1}^{n} v_{xi} - 1\right) - A\left(\sum_{y=1}^{n} v_{yi} - 1\right) - D\sum_{y=1}^{n} d_{xy}\, v_{y,i+1} \qquad (5)$$

where, *n* represents the number of neurons in the Hopfield network. $v_j(t)$ and $U_j(t)$ are the voltages of the output and input, respectively. The dynamic equation of the Hopfield network for TSP is acquired as follows. In the initialization stage,

$$U_{xi}(t) = U_0 \ln(N-1) + \delta_{xi} \quad (x, i = 1,2,\ldots,N; t = 0) \quad (6)$$

where, $U_0 = 0.1$; *N* is the number of waypoints; δ_{xi} is a random value ranged at (-1,+1). The constants *A* and *D* are selected as *A* = 200, and *D* = 100. Step length is 0.0001.

The following is the steps of TSP computation for multi-goal visit.

Step 1 Input: coordinates of *N* waypoints. Computation of the distance between *N* waypoints;

Step 2 Initialization of the Hopfield neural network;

Step 3 Computation $\frac{dU_{xi}}{dt}$ of based on Eq. (5). The following is calculated on the basis of Euler method.

$$U_{xi}(t+1) = U_{xi}(\mathrm{t}) + \frac{dU_{xi}}{dt}\Delta T \quad (7)$$

Step 4 Computation of $v_{xi}(t)$ by the following equation.

$$v_{xi}(t) = g\big(U_{xi}(t)\big) = \frac{1}{2}\left[1 + tansig\left(\frac{U_{xi}(t)}{U_0}\right)\right] \quad (8)$$

Step 5 The energy function *E* is obtained by Equs. (2) and (3).

Step 6 Iteration is done. $k > 10{,}000$? Otherwise, $k = k + 1$, go to Step 3.

After execution of this HNN-based TSP algorithm, the minimized total route to connect all the waypoints is obtained. There are 15 waypoints in the workspace that required a service mobile vehicle to reach per waypoint. The HNN-based TSP algorithm is applied to this application illustrated in Fig. 2.

IV. Real-Time Concurrent Map Building and Multi-Goal Navigator

Given a set of waypoints as multiple goals and possible starting points, a matching lookup table for waypoint sequencing is created. The D*-Lite algorithm then provides an initial path marked by markers between pairs of goals, and the VFH navigation algorithm drives the vehicle alone those markers. Path planning is carried out using the D*-Lite algorithm, which provides the best route between waypoints. D*-Lite can work off a partially complete map of the field, and progressively re-plans the optimal route when the map is augmented with new information as the vehicle explores.

Although D*-Lite is a very effective algorithm to use for unknown terrain, its integration with a specific navigation task is difficult without a built-in debugging tool. Therefore, we designed a debugging tool which shows the markers path from one waypoint to another as D*-Lite plans. Our program

prints out these markers, which update continuously according to the movement of the vehicle.

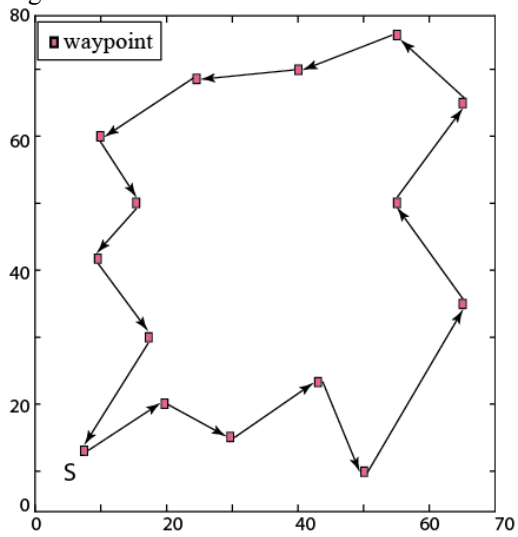


Fig.2 HNN algorithm for TSP applied in a 15-waypoint workspace

A sort of multi-layer software development environment is implemented based on Player/Stage, an open-source, Unix-based robotic simulation software system, in which the proposed algorithms are developed, debugged, tested and simulated. Player provides an interface for a variety of vehicle and sensor peripherals accessed from local computers via a network connection to the vehicle whereas Stage™ is a simulation system for a vehicle and its environment. Since Stage™ can promptly simulate, validate and test a developing algorithm under an off-vehicle environment, it usually acts as a simulator with the vehicle, obstacles and sensors (such as LIDAR, camera, digital compass and GPS), to accelerate the development procedure of a new algorithm. Stage™ provides a powerful simulation environment that can be employed to develop and test algorithms in environments similar to those expected in an actual autonomous vehicle.

These multiple layers on Player/Stage provide modular isolation and enable seamless communication between software and either actual hardware or a hardware simulator possible. The first is the server layer which is occupied by Player, an open-source, Unix-based robotic software system that serves as an interface between robotic algorithm implementations and the vehicle systems. Specifically, it provides standard interfaces for a typical set of robotic peripherals (LIDAR, cameras, motors, etc.). The second layer is the client layer occupied by user programs written in MATLAB. These layers communicate with Player over a TCP socket to acquire data from sensors and send actuator commands, which Player then passes on to the vehicle. An actual autonomous vehicle was developed pictured in Fig. 3 as test-bed for our hybrid system of real-time concurrent multi-goal navigation and map building of an autonomous vehicle. The vehicle incorporates six sensors into its compact design as follows: a LIDAR, a DGPS, a digital compass, a camera, and an IMU, each of which is enclosed in a waterproof case and firmly mounted to the vehicle.

A 270^0 SICK LMS111 LIDAR is configured for the purposes of obstacle detection illustrated in Fig.3. The unit is able to obtain data over a 270° field-of-view with 0.25° resolution, a maximum range of 20 m, and a 25Hz scanning rate. The sensor configuration of our autonomous vehicle for multi-goal navigation and mapping is illustrated in Fig. 3.

V. Simulation and Experiments

In this section, the proposed hybrid system for real-time concurrent multi-goal navigation and mapping of an autonomous vehicle in unknown environments will be validated first on Player/Stage simulator and then on an actual autonomous vehicle.

In order to validate the effectiveness and efficiency of the proposed hybrid system, the model is applied to simulate such real-time concurrent multi-goal navigation and map building of an autonomous vehicle under unknown environments in a thirty-goal course.

In this simulation, the vehicle is navigated to connect 15 waypoints in a free-space to test the HNN-based multi-goal model with mapping at the early stage illustrated in Fig. 4.

The vehicle is able to traverse from the initial point to plan the shortest route to visit the 15 waypoint while the vehicle builds the map with 270 LIDAR. The rout of multi-goal by the vehicle is shown in Fig. 4(a) whereas the map built is illustrated in Fig. 4(b) at the end of the travel of the vehicle. In this simulation, the vehicle is guided in an *unknown* environment populated with obstacles with 15 waypoints as multiple goals depicted in Fig. 4, which shows that the vehicle traverses from starting waypoint to connect 15 waypoints obtained by GPS coordinates.

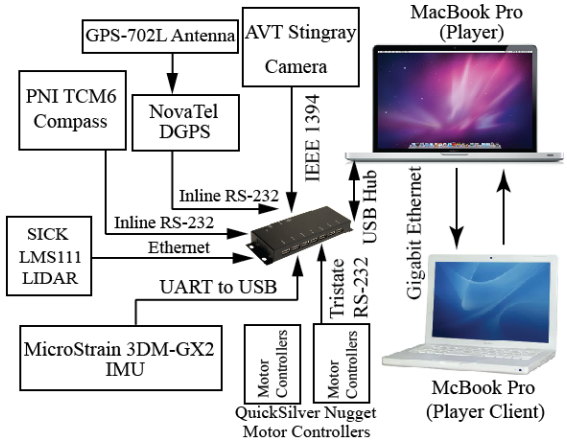
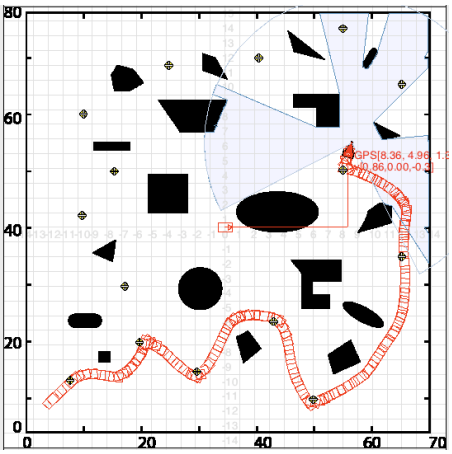


Fig. 3 The sensor configuration of our autonomous vehicle for multi-goal navigation and mapping

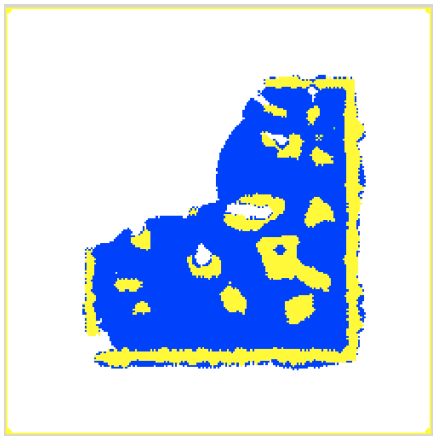
The built map while the vehicle moves in the unknown environment with 270° LIDAR scan is illustrated in Fig. 4(a) while it moves on the road from waypoint to waypoint. The yellow portions are detected obstacles in Fig. 4(b).

In accordance with the planed trajectory, the GPS coordinates of the targets in latitude and longitude in sequence of the goals is dispatched to the D*-Lite based global path planner. As described in the previous sections, from one goal to another goal, the VFH based local navigator

is utilized to navigate the vehicle by following pre-placed markers. Starting from the initial point, the autonomous vehicle is capable of travelling to the sequence of goals based on the proposed multi-goal navigation system and experiment was successfully accomplished.



(a)



(b)

Fig 4. Simulation result of the vehicle in unknown workspace with 15-waypoint by the proposed HNN-based TSP model at the early stage of movement. (a): The workspace with obstacles and 15 goals; (b): Built map.

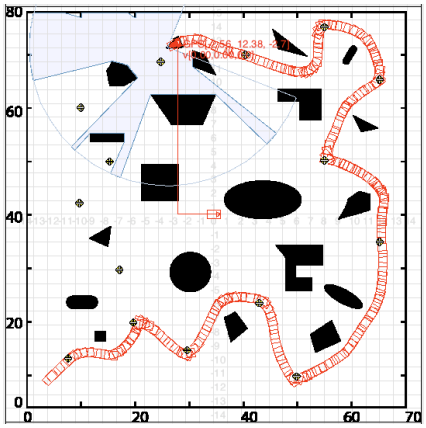
In the experimental test field, the vehicle is required to autonomously travel from a predefined starting point to a number of target destinations as waypoints and move back to home base, in which the waypoints are given only by the GPS coordinates of the targets in latitude and longitude.

After the route to visit these fifteen waypoints is generated by the proposed HNN-based TSP algorithm, the GPS coordinates of the targets in latitude and longitude in sequence of the goals are transmitted to the global path planner in Fig. 5. Once a path from D* Lite has been obtained that was early generated by the HNN-based TSP algorithm, a number of points along it are extracted so as to use the VFH algorithm between two goals. These points as markers are converted into GPS coordinates and presented to VFH as consecutive points. VFH then generates motion commands which are transmitted to the drive controllers to move the vehicle towards these intermediate waypoints [14]. The planned route and map built of the vehicle in unknown workspace with 15-waypoint at the middle stage of movement are illustrated in Fig. 5(a) and Fig. 5(b), respectively.

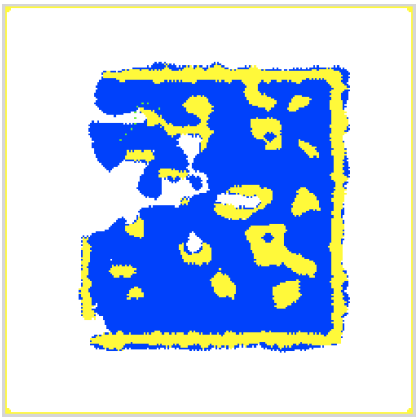
Once the vehicle approaches to an intermediate waypoint, the next intermediate goal along the desired path is regarded to be achieved and substituted. The planned route and map built of the vehicle in unknown workspace with 15-waypoint while it has traverses the most region are illustrated in Fig. 6(a) and Fig. 6(b), respectively.

In Fig. 7, the autonomous vehicle returns the starting point after it traverses from starting waypoint to every goal by planning a collision-free trajectory to connect these thirty goals with minimized overall distance.

Once the vehicle approaches the initial waypoint, the all the waypoints are visited once. The planned route and map built of the vehicle in unknown workspace with 15-waypoint at the end are illustrated in Fig. 7(a) and Fig. 7(b), respectively. The developed actual autonomous vehicle is imaged in Fig. 8 for the test purpose.



(a)



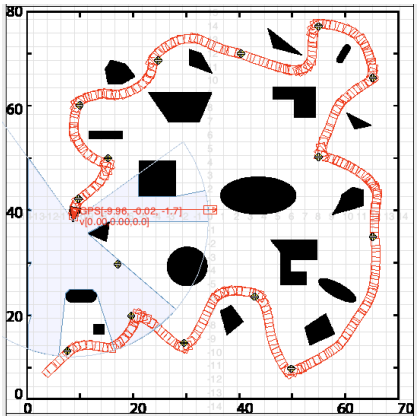
(b)

Fig 5. Simulation result of the vehicle in unknown workspace with 15-waypoint at the middle stage of movement by the proposed HNN-based TSP model. (a): The workspace with obstacles and 15 goals; (b): Built map.

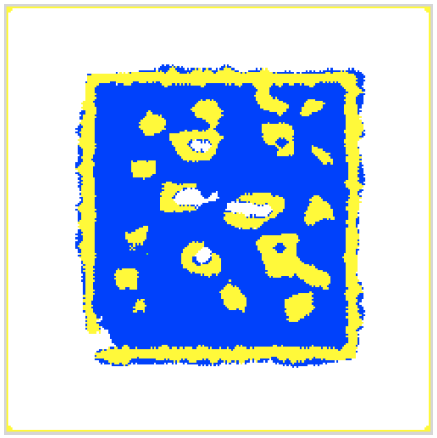
VI. Conclusion

A solution by hybrid algorithms was developed in this paper for real-time map building and navigation for **multiple goals** purpose. In this paper, an alternate approach, the D*-Lite algorithm associated with a local LIDAR-based

navigation methodology was developed for multiple goals. The D*-Lite path planning algorithm was used to provide VFH with intermediate goals. The multi-goal route was calculated and planned by the proposed HNN-based TSP strategy. Results from simulation studies and experiments demonstrated the benefits of the local navigator in conjunction with a path planner to reach multiple goals with minimized total distance.

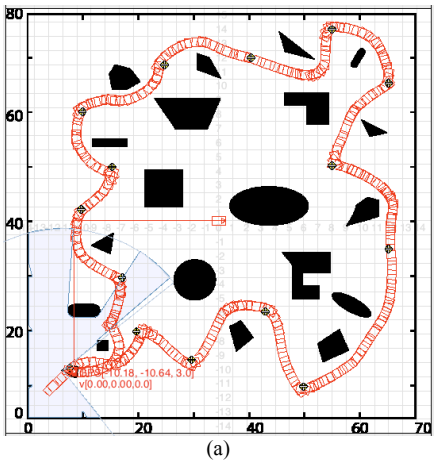


(a)

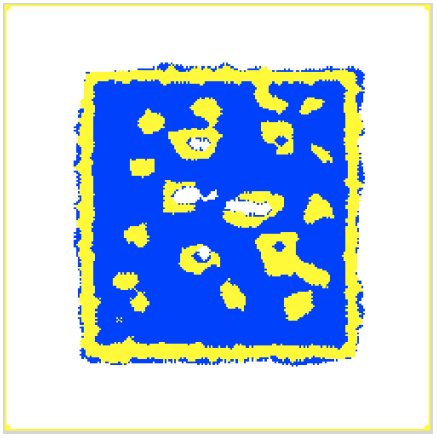


(b)

Fig. 6 The route planned by the proposed HNN-based TSP model during the trip of the vehicle (a): The workspace with obstacles and 15 goals; (b): Built map.



(a)



(b)

Fig 7. Simulation result of the vehicle in unknown workspace with 15-waypoint by the proposed HNN-based TSP model at the end. (a): The workspace with obstacles and 15 goals; (b): Built map.

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